How to Avoid Flight Delays When Flying from New York City

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```r
# Load any R Packages you may need
library(tidyverse)
library(nycflights13)
library(plotly)
library(mosaic)
```

Introduction

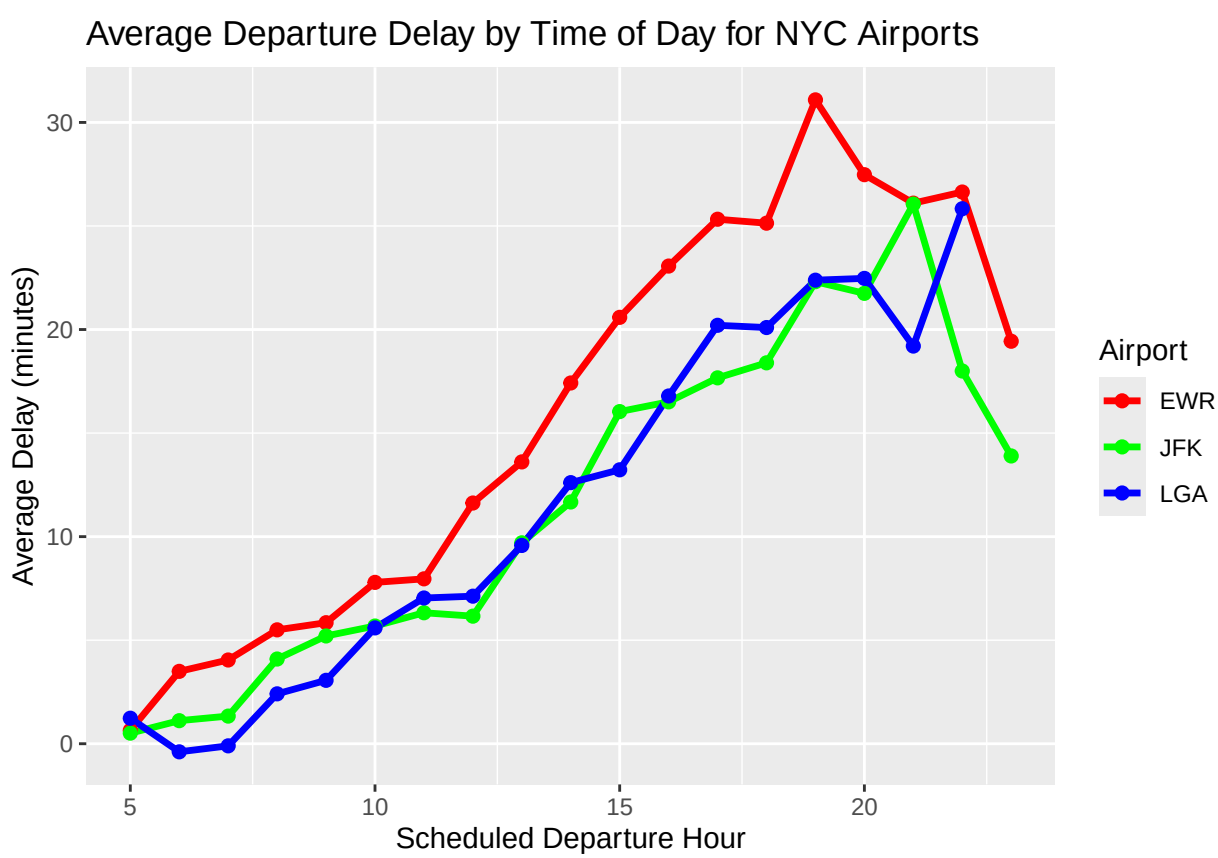
Flight delays are very common, especially at busy airports like those in New York City. But delays do not happen randomly. Some flights are much more likely to be delayed than others. In this project, we use flight data from 2013 to see when delays happen most often and which airport performs best. This information can help travelers choose flights that are less likely to leave late.

Dataset Description

This project uses the nycflights13 dataset. It includes information about every commercial flight that left New York City in 2013 from JFK, LaGuardia (LGA), and Newark (EWR). The data contains departure times, delays, airlines, airports, and weather details. Because the dataset is large and detailed, it is useful for studying patterns in flight delays.

NYC Airports Delays Throughout the Day

```r
flights %>%
  filter(!is.na(dep_delay)) %>%
  group_by(origin, hour) %>%
  summarize(avg_delay = mean(dep_delay), .groups = "drop") %>%
  ggplot(aes(x = hour, y = avg_delay, color = origin)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  labs(
    title = "Average Departure Delay by Time of Day for NYC Airports",
    x = "Scheduled Departure Hour",
    y = "Average Delay (minutes)",
    color = "Airport"
  ) +
  scale_color_manual(values = c("EWR" = "red", "JFK" = "green", "LGA" = "blue")
                     )
```



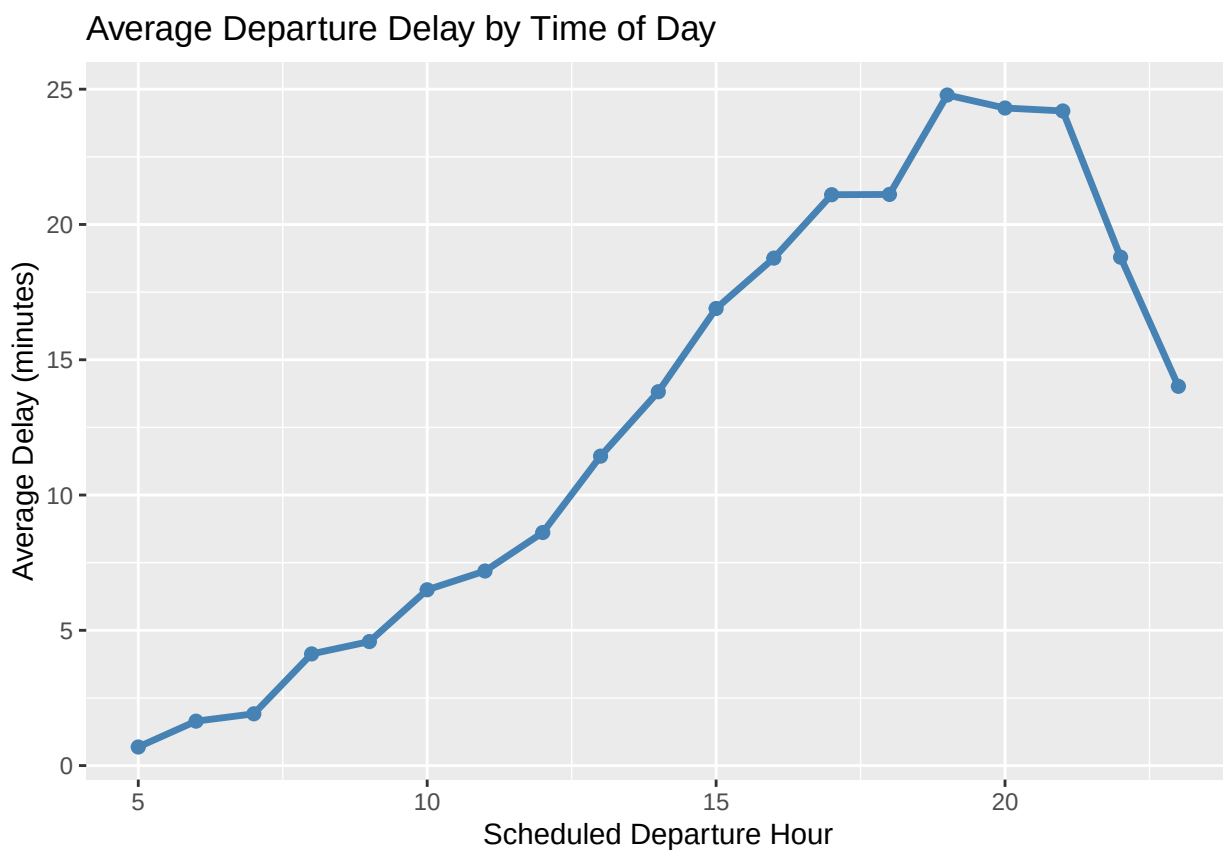
Interpretation

The visualizations suggest when analyzing departure delays by hour for each NYC airport, clear differences emerge. Newark (EWR) consistently shows the highest delays, especially in the afternoon and evening, indicating that delays accumulate more severely there and then at the end of the day it goes back down. JFK displays a moderate pattern: morning flights are generally on time, but delays increase steadily throughout the day and gets a spike at the end of the day and goes back down. LaGuardia (LGA) performs the best overall, with the lowest average delays and a more stable pattern across hours. However, the interesting thing is that as both EWR and JFK average delay time increases and then starts to decrease by the end of

the day.But LGA does the opposite, the average delay time suddenly decreases and then starts to spike at the end of the day. These results suggest that travelers seeking to minimize delay risk should choose morning flights and, when possible, depart from LGA rather than EWR. However, if choosing a night flight, I would recommend JFK as it has the least delay time.

Overall Delays Increase Throughout the Day

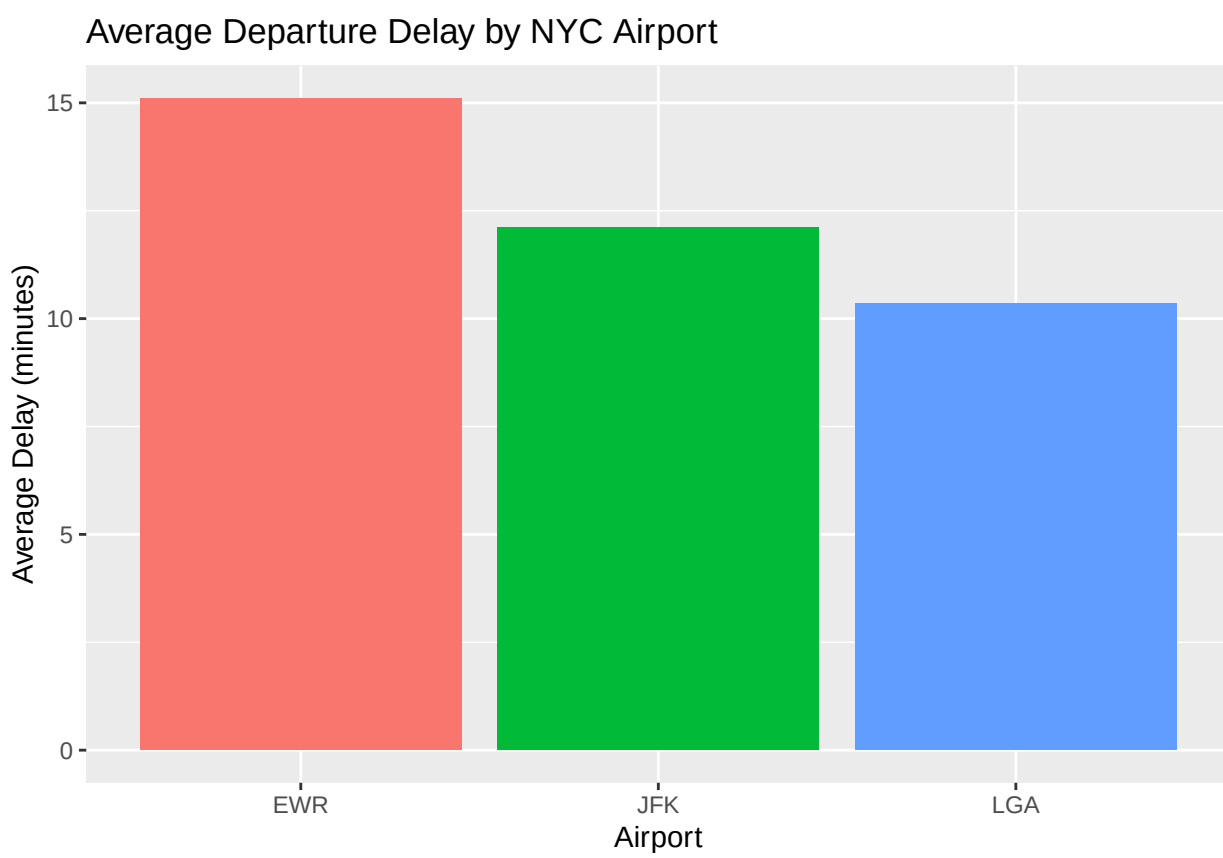
```
flights %>%
  filter(!is.na(dep_delay)) %>%
  group_by(hour) %>%
  summarize(avg_delay = mean(dep_delay)) %>%
  ggplot(aes(x = hour, y = avg_delay)) +
  geom_line(color = "steelblue", linewidth = 1.2) +
  geom_point(color = "steelblue", size = 2) +
  labs(
    title = "Average Departure Delay by Time of Day",
    x = "Scheduled Departure Hour",
    y = "Average Delay (minutes)"
  )
```



Flights departing early in the morning tend to leave on time, with very small average delays. However, delays increase steadily throughout the day and are highest in the evening. This pattern suggests that delays accumulate as the day progresses. When earlier flights arrive late, later departures are affected, creating a chain reaction that leads to larger delays later in the day.

Which NYC Airport Is Most Reliable?

```
flights %>%
  filter(!is.na(dep_delay)) %>%
  group_by(origin) %>%
  summarize(avg_delay = mean(dep_delay)) %>%
  ggplot(aes(x = origin, y = avg_delay, fill = origin)) +
  geom_col(show.legend = FALSE) +
  labs(
    title = "Average Departure Delay by NYC Airport",
    x = "Airport",
    y = "Average Delay (minutes)"
  )
```



The three New York City airports do not have the same average delays. Newark (EWR) has the largest average delay, while LaGuardia (LGA) has the smallest. JFK falls in between the two. These differences may be caused by airport traffic, runway availability, or how busy each airport is. Choosing an airport with smaller delays can improve the chances of leaving on time.

Conclusion: Tips for Avoiding Delays

The data shows a simple way to reduce the chance of delays:

-Choose flights early in the morning

-Pay attention to which airport you use

-Avoid flights late in the day if possible

No flight is guaranteed to be on time, but these choices can help lower the risk. Using data like this can make travel planning easier and less stressful.

References

dataset information: https://github.com/tidyverse/nycflights13/blob/main/README.md

ggplot cheatsheet: https://posit.co/wp-content/uploads/2022/10/data-visualization-1.pdf